Unlocking
Customer Behavior

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Central Idea

In the fast-paced world of online marketing and offline retail, understanding the intricate factors that drive customer behavior and provide detail service are essential for retail businesses to stay competitive.

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Data Description

- Dataset Overview: The dataset provides the consumer purchasing behavior and customer feedback for retails (reference from Kaggle : Retail Analysis Large Dataset)
- Dataset Size: Over 300,000 transactions

Data Dictionary

Variable	Description	Data Type	Example Values	Unit
`Transaction_ID`	Unique identifier for each transaction	Numeric	`123456`	None
`Customer_ID`	Unique identifier for each customer	Numeric	`78910`	None
`Product_Category`	Product category indicating the primary type of product purchased	Categorical	`Electronics`, `Clothing`, `Home Decor`	None
`Product_Brand`	Brand of the product	Categorical	`Nike`, `Samsung`	None
`Shipping_Method`	Shipping option selected for the product	Categorical	`Same-Day`, `Standard`	None
`Feedback`	Customer feedback indicating satisfaction level	Categorical	`Excellent`, `Average`, `Bad`	None
`Total_Amount`	Total amount spent by the customer during the transaction	Numeric	`250.50`	USD
`Ratings`	Customer rating of the product, on a scale of 1 to 5	Numeric	`4.5`	None
`Age`	Customer's age	Numeric	`28`	Years
`Gender`	Customer's gender	Categorical	`Male`, `Female`	None
`Feedback_Count`	Total number of feedbacks provided by the customer for a product	Numeric	`3`	Count

Data Preparation

Handing missing values

Numerical Columns:
Filled with the median to minimize distortion.

Categorical Columns:
Filled with the mode to ensure logical consistency.

Remove Duplicates

Eliminated repeated rows to maintain data integrity.

Standardize Categorical Values

Removed leading/trailing spaces.
Standardized capitalization

Remove Outliers

Applied Interquartile Range (IQR) method.

Focused on middle 50% of data to eliminate anomalies.

Normalize Column Names

Converted names to **lowercase**.

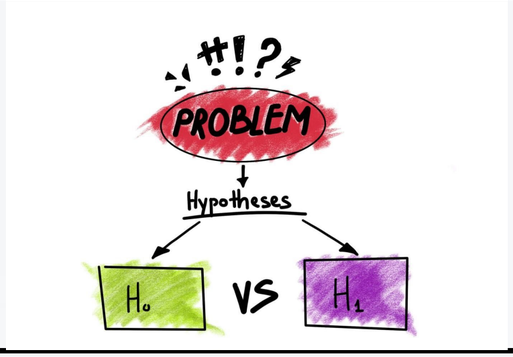
Replaced spaces with underscores for easier navigation.

HYPOTHESIS 1

Age groups influence total purchase amount

Hypothesis 1

Hypothesis



Older consumers are more inclined to purchase high-quality products with high ratings to meet their family life and health needs, while younger consumers are influenced by social media, promotions, and user reviews and tend to choose trendy products with high ratings.

Key Variables

- Age : divide in age groups (1"18-25", "26-35", "36-45", "46-55", "56-65", "66-70")
- High-rated products : filter rating over 4.5
- Total purchase amount

Analysis Methods

Find the relationship between key variables with visualization

Data Exploration

Using Chi-square test to determine the relationship between age groups' purchase amount with high-rated products, visualization

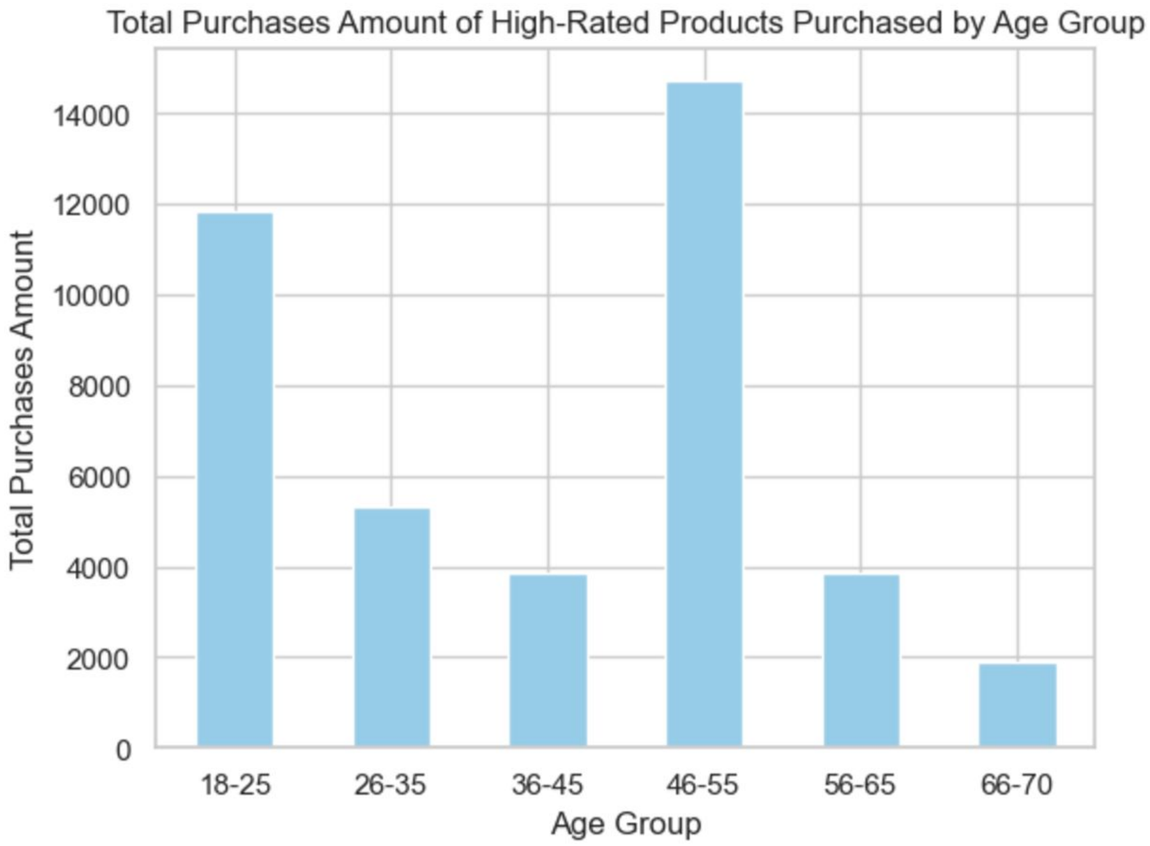
Statistical Test

Using decision tree and random forest regression model to verify the specific high-rated products' purchase amount among the age groups

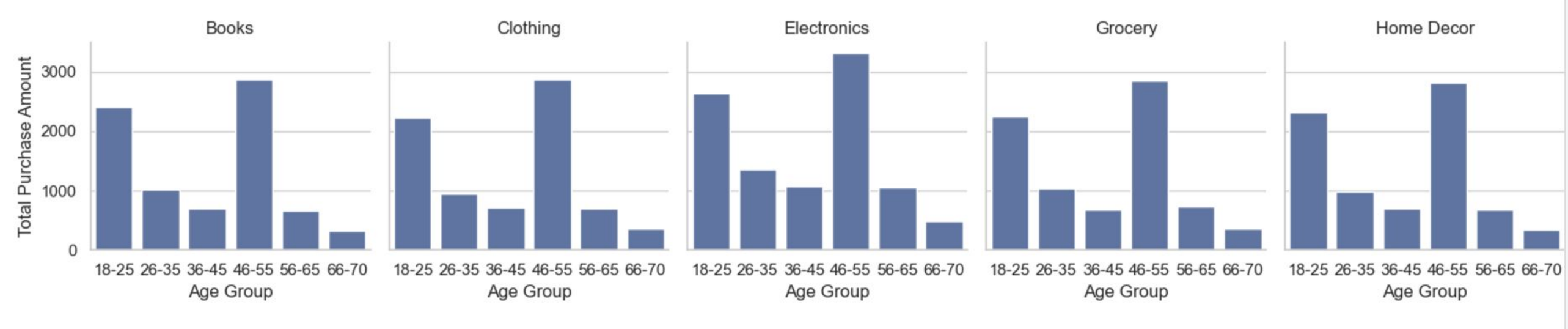
Further Analysis

the relationship between the number of purchases from different age groups

- Age group between 46-55 reached the highest purchase amount
- Age group between 18-25 follow them as the second group
- The oldest age group has the lowest purchase amount



the relationship between the number of purchases from different age groups in specific product categories



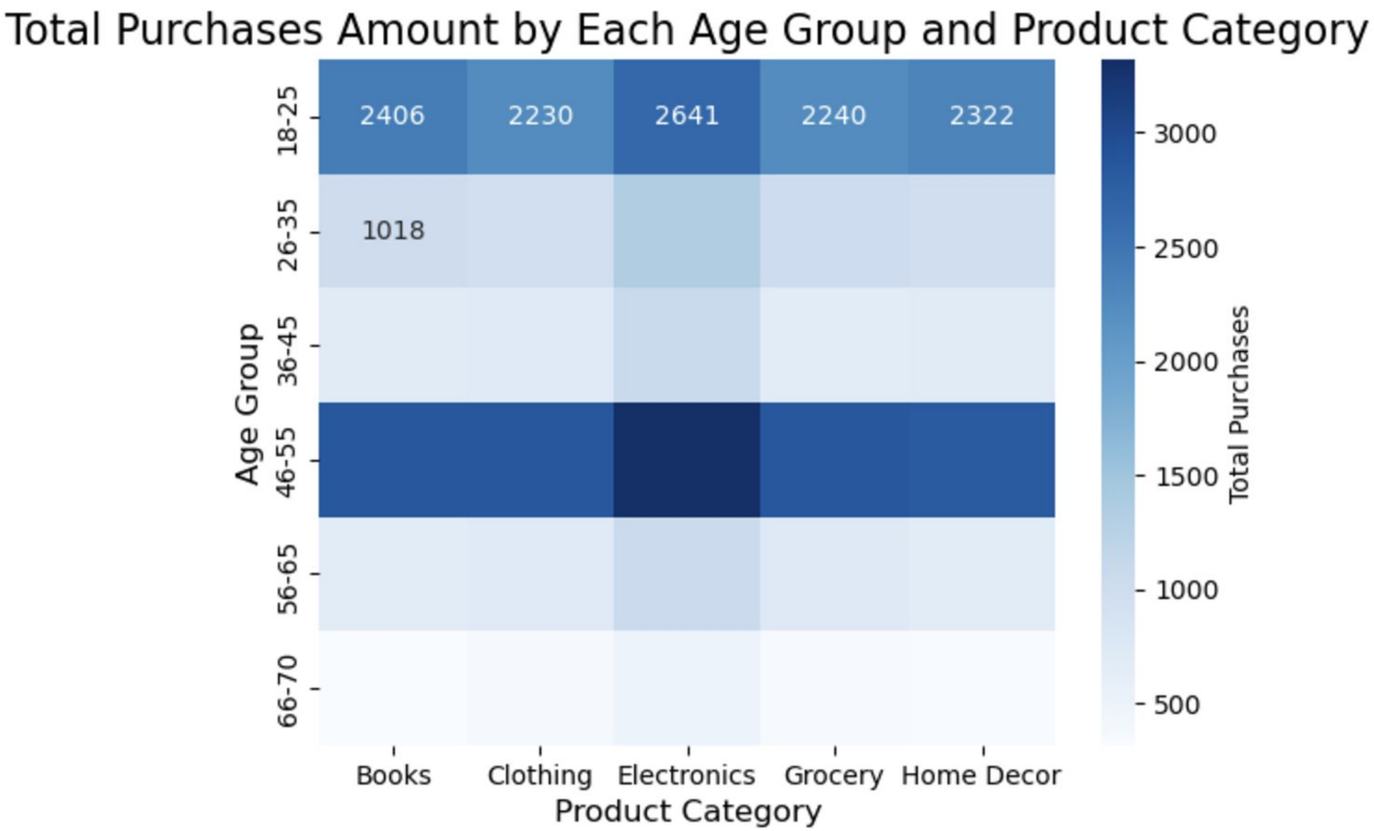
- For all categories, age group 46-55's purchase amount reached the highest and 18-25 occupied the second position
- For the other age groups' change is not obviously

Chi-square test

- Chi-value :121.1139
- P-value: 1.7726910985010154e-16
- Conclusion: There is a significant difference

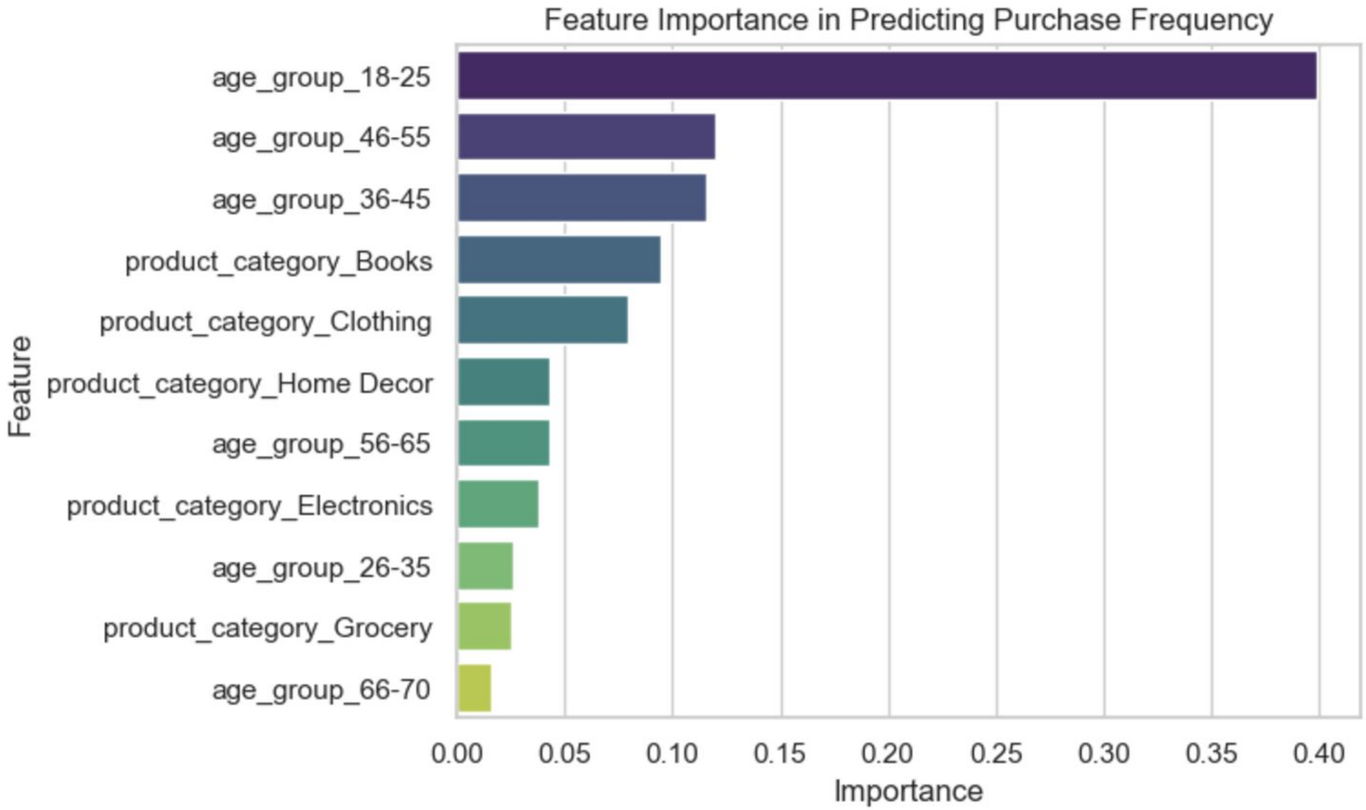
  in purchase amount of high-rated products

  between age groups

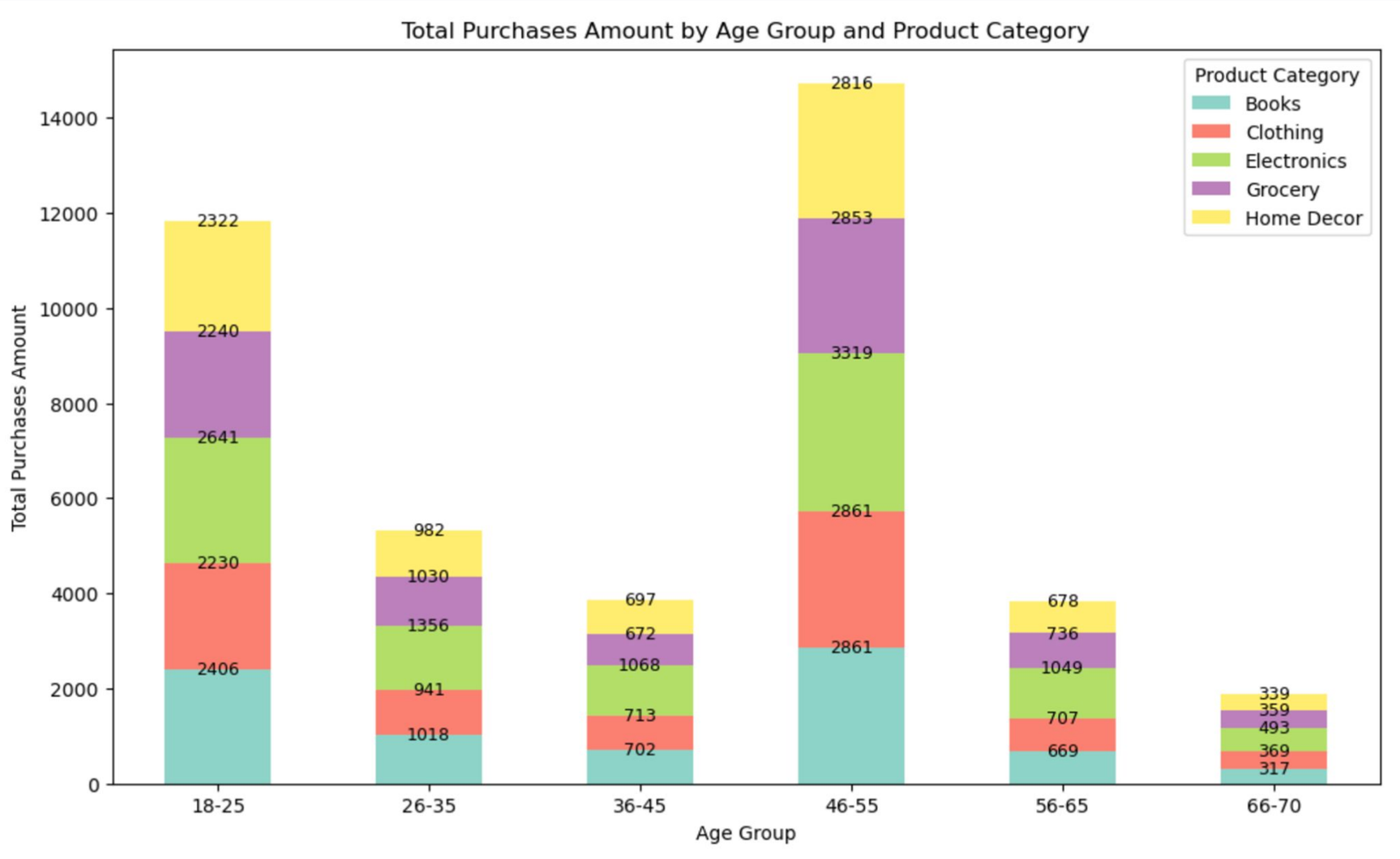


Further Analysis - Decision Tree

- X = filtered_data[['age_group', 'product_category']]
- y = filtered_data['total_purchases']
- the two groups of 18-25 and 46-55 years old, which shows that age has a significant impact on purchasing behavior

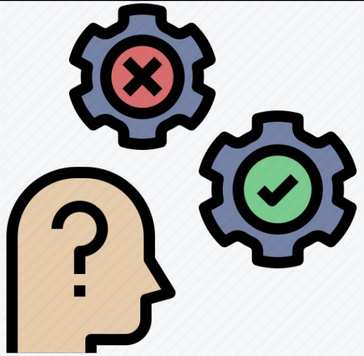


Further Analysis - Random Forest Regression Model



Insights

Conlusion



- Age is strongly associated with the total purchase amount of highly-rated products: older consumers are more likely to buy high-quality, well-rated items to fulfill their family and health needs, while younger consumers are influenced by social media, promotions, and user reviews, often opting for trendy, highly-rated products.

- Retailers can adopt age-targeted strategies by offering trendy products to younger consumers and emphasizing highly-rated, high-quality items for older customers. Focusing on top-rated products, enhancing quality, and improving customer service will strengthen competitiveness, boost customer loyalty, and drive sales growth.

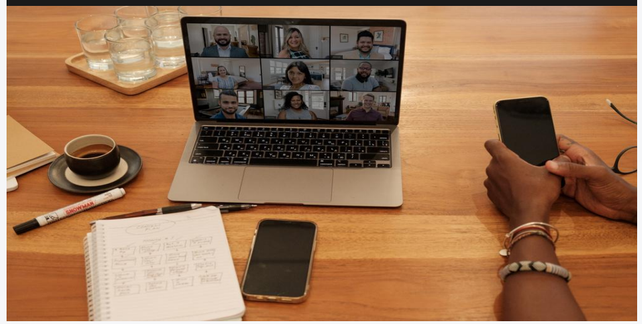
Insights

HYPOTHESIS 2

Purchase decision differs between male and female

Hypothesis 2

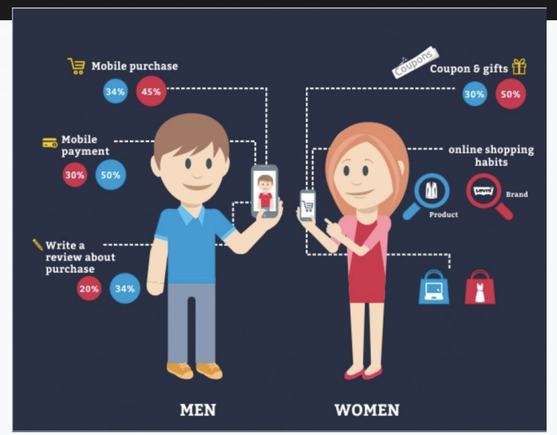
Hypothesis



Men prioritize functionality and technology, spending more on high-value items, while women focus on lifestyle and aesthetics, often aligning their purchases with value-driven choices.

- Gender
- Total amount
- Product category
- Product brands

Key Variables



Analysis Methods

Conduct a t-test and chi-squared test to assess the relation between genders and spending amount, product category and product brand

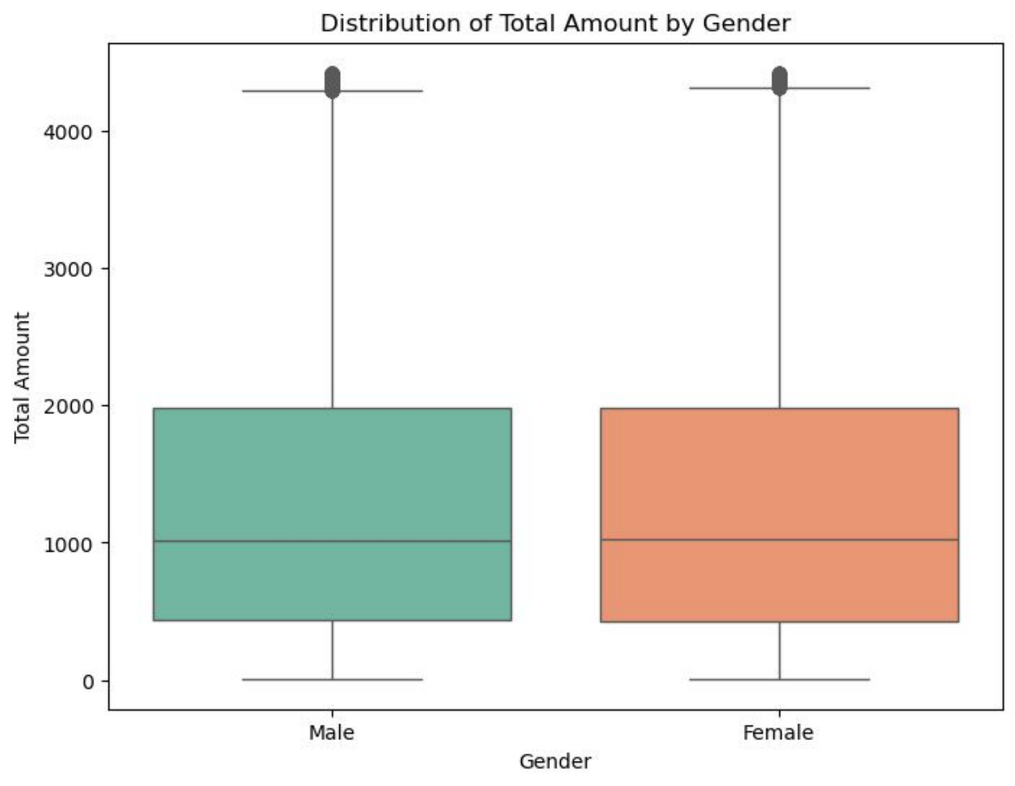
Statistical Test

Further Analysis

Perform a multinomial regression analysis to examine the relationship and use classification to predict the gender

Machine Learning

T-test - The difference in spending across genders

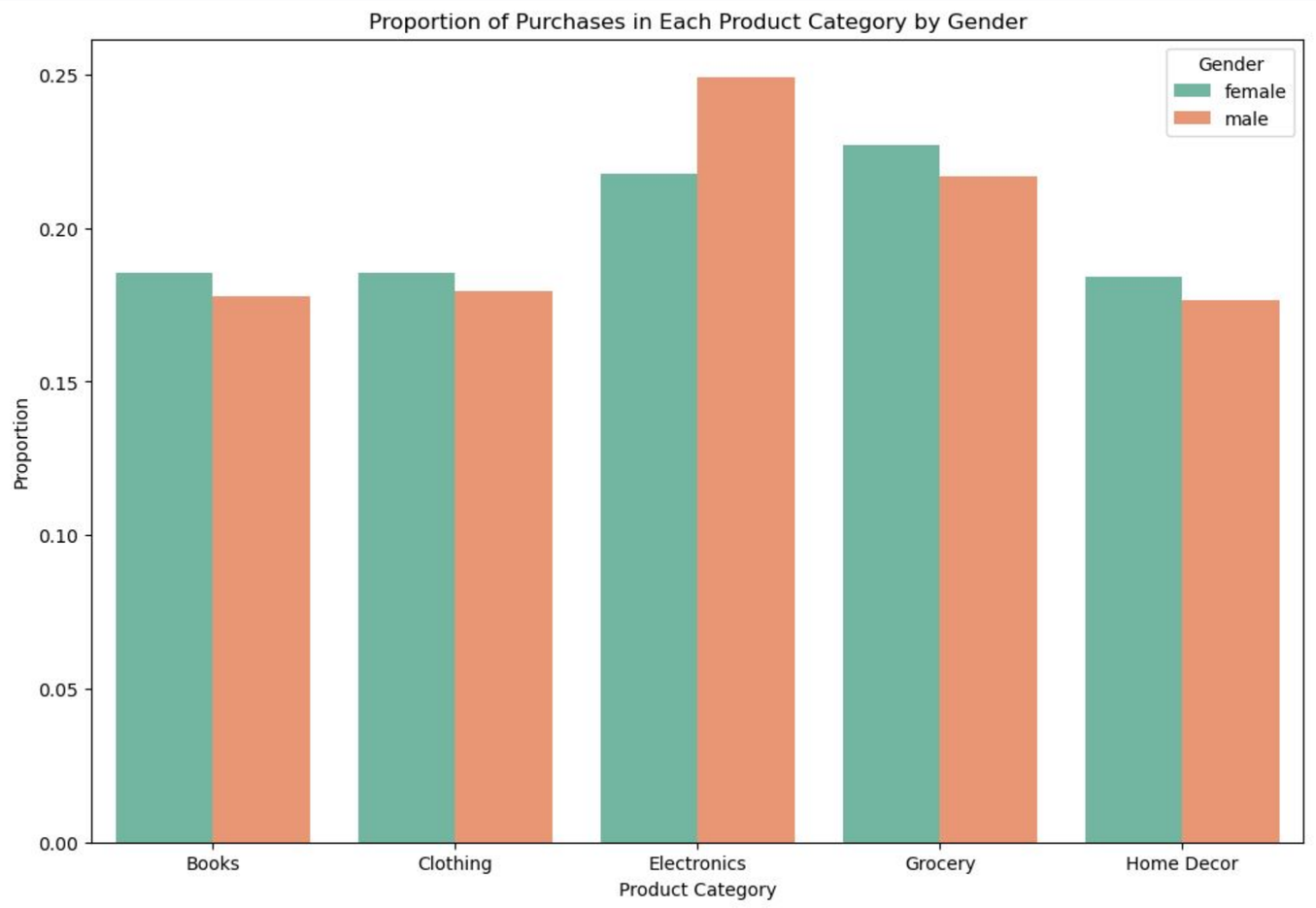


Male Spending Statistics:
Max: 4421.99
Min: 10.00
Average: 1324.15
Median: 1020.77

Female Spending Statistics:
Max: 4422.00
Min: 10.06
Average: 1326.74
Median: 1024.03

- T-test result: T-statistical=-0.5854, p=0.5583
- The median consumption of men and women is similar, but the upper and lower limits of women's consumption data are slightly wider, which means that the female consumer group is more widely distributed.
- There are a few outliers with high consumption values for both men and women.

The difference in Product Categories across genders

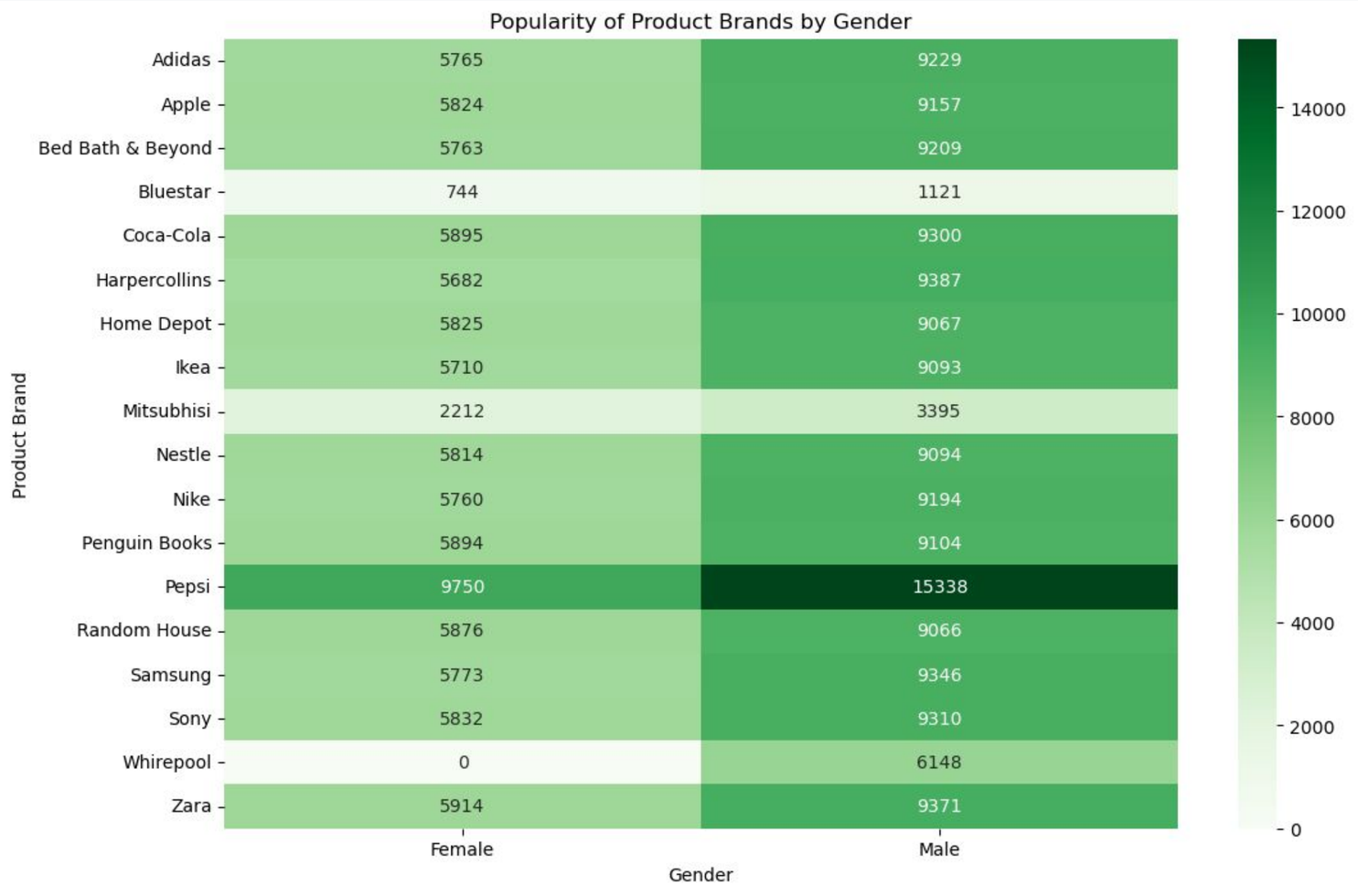


Category	Gender	Books	Clothing	Electronics	Grocery
Expected	Female	17002.93	18078.25	22323.98	20751.23
	Male	28014.07	28144.75	36781.03	34189.78

Category	Gender	Books	Clothing	Electronics	Grocery
Proportion	Female	0.185	0.185	0.218	0.227
	Male	0.178	0.179	0.249	0.217

- Chi-Squared Statistic: 322.7654, p=0.0000,
- Degrees of freedom = 4
- Men have significantly higher preferences for electronic products and household items than women.
- Women's preferences are more concentrated in book and clothing categories.

The difference in Product Brand across genders



- Chi-Squared Test Results:
- Chi-Squared Statistic: 3844.5
- P-Value: 0.00
- Degrees of Freedom: 17
- Male users have a significantly higher preference for beverage brands (such as Pepsi) than female users.
- There is little difference in preference between men and women for some common brands (such as Apple and Nike).

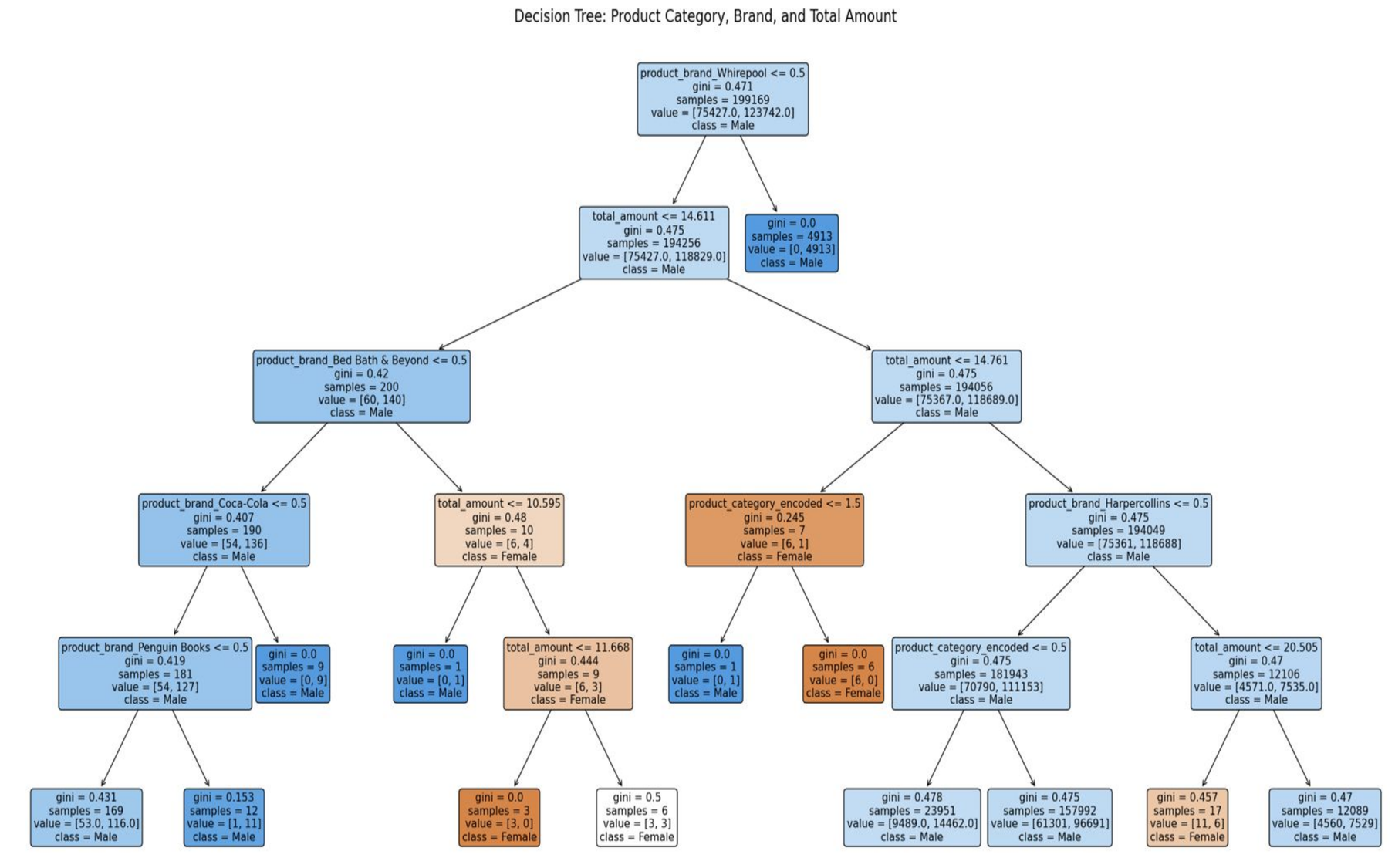
Further Analysis—Logistic Regression

- It was found that men prefer electronic products (such as Samsung, Whirlpool), while women prefer books and clothing (such as Blue Star, Zara).
- The results quantify significant gender differences in product and brand choices

Class	Coefficient	Intercept
Books	-0.0372	-0.0703
Clothing	-0.0281	-0.0710
Electronics	0.1420	0.0886
Grocery	-0.0410	0.1320
Home Decor	-0.0358	-0.0790

Brand	Coefficient	Intercept
Zara	-0.3512	0.5347
Bluestar	-0.3941	-1.5440
Whirlpool	6.0249	-6.2639
Samsung	-0.3281	0.5098

Machine Learning - Classification



- Target variable: Gender
- Feature variables: Total amount, Product categories, Product brands
- Whirepool is the top split
- Total purchase amount strongly influences
- Electronic: Male oriented
- Books and Home Decor: Female aligned
- Gini impurity: The tree performs well generally, but some variables are less decisive

INSIGHTS

Conclusion

Gender affects spending behavior, brand preferences, and product category choices, with men favoring electronics \while women lean towards books, clothing, and home decor.

- Targeted Marketing
- Pricing Stratey
- Product Placement
- Inventory Management

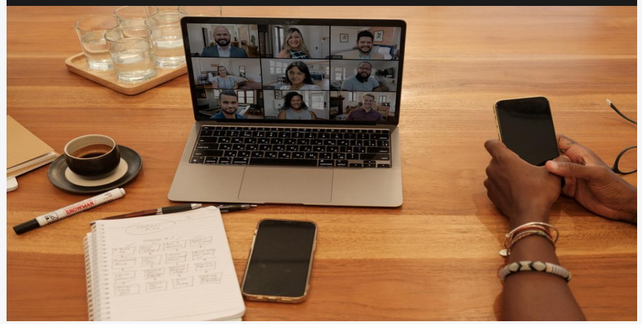
Insights

HYPOTHESIS 3

Shipping methods and customer feedback

Hypothesis3

Hypothesis



Same-Day Shipping" is associated with significantly higher feedback scores, especially in time-sensitive categories such as "Electronics" and "Clothing.

Standard Shipping" shows weaker impacts on feedback, with higher likelihoods of neutral or negative feedback, particularly in less time-sensitive categories such as "Home Decor.

- Customer Feedback Scores (Excellent, Good, Average, Bad)
- Product Categories (Electronics, Clothing, Home Decor)
- Shipping Methods (Same-Day, Standard)

Key Variables

Analysis Methods

classification model

Evaluate model performance (precision, recall, ROC-AUC) and analyze feature importance, focusing on demographic and interaction variables.

ANOVA

Compare feedback scores across "Same-Day" and "Standard" shipping to assess the impact of shipping speed.

Tukey HSD

Quantify pairwise differences in feedback scores between shipping methods.

Chi-Square Test

Examine the association between shipping methods and feedback categories.

Multi-Category Regression

OLS regression identified key predictors and interaction effects on feedback scores.

Cramér's V

Measure relationship strength and analyze effect sizes for time-sensitive products.

Visualization

Use box plots and bar charts to illustrate feedback distribution and shipping method impact.

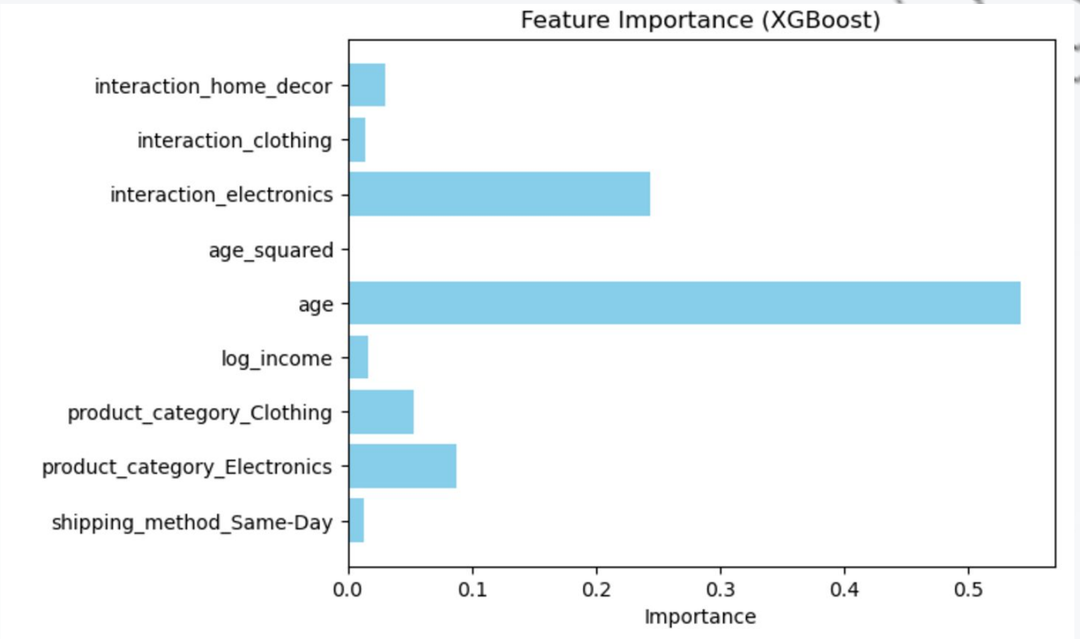
Classification Model

```
Classification Report:
              precision    recall  f1-score   support

           0       0.73      0.94      0.82     49821
           1       0.72      0.29      0.41     24868

    accuracy                           0.73     74689
   macro avg       0.72      0.62      0.62     74689
weighted avg       0.72      0.73      0.69     74689

ROC-AUC Score: 0.6955282245165051
```



Performance:

- ROC-AUC: **0.695**, indicating moderate model performance.
- Recall for "Excellent" feedback: **0.29**, showing difficulty in identifying positive cases.

Key Insights:

- Age and interaction terms (e.g., interaction_electronics) are the most influential predictors.
- Highlights demographic effects and the significance of time-sensitive products.

ANOVA and Tukey HSD Analysis

```
ANOVA Results:
 F_onewayResult(statistic=101.26180373352278, pvalue=8.144542537112282e-24)

Post-hoc Tukey HSD:
 Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
group1 group2 meandiff p-adj lower  upper  reject
-----------------------------------------------------
 False   True   0.0442   0.0 0.0356 0.0528   True
-----------------------------------------------------
```

ANOVA:

- Significant differences in feedback scores between "Same-Day" and "Standard" shipping methods.
- F = **101.26**, p < **0.001**.

Tukey HSD:

- Confirmed a statistically significant mean difference of **0.0442** favoring "Same-Day" shipping.

Multi-Category Regression

```
Regression Results:
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.058
Model:                            OLS   Adj. R-squared:                  0.058
Method:                 Least Squares   F-statistic:                     1700.
Date:                Wed, 04 Dec 2024   Prob (F-statistic):               0.00
Time:                        18:26:05   Log-Likelihood:            -3.5646e+05
No. Observations:              248962   AIC:                         7.129e+05
Df Residuals:                  248952   BIC:                         7.130e+05
Df Model:                           9
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.4317      0.018     23.768      0.000       0.396       0.467
x1             0.0563      0.005     10.573      0.000       0.046       0.067
x2             0.0983      0.005     18.680      0.000       0.088       0.109
x3             0.0037      0.007      0.540      0.589      -0.010       0.017
x4             0.0162      0.007      2.326      0.020       0.003       0.030
x5            -0.1241      0.008    -15.974      0.000      -0.139      -0.109
x6             0.0743      0.001     88.090      0.000       0.073       0.076
x7            -0.0008   1.03e-05    -74.504      0.000      -0.001      -0.001
x8            -0.0488      0.011     -4.276      0.000      -0.071      -0.026
x9            -0.0566      0.011     -4.950      0.000      -0.079      -0.034
==============================================================================
Omnibus:                    38870.424   Durbin-Watson:                   1.421
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            14559.981
Skew:                          -0.390   Prob(JB):                         0.00
Kurtosis:                       2.108   Cond. No.                     1.79e+04
==============================================================================
```

Significant Predictors:

- Same-Day Shipping (x1): Positive, $p<0.001$
- Electronics (x2): Positive, $p<0.001$
- Age (x6): Strong positive, $p<0.001$

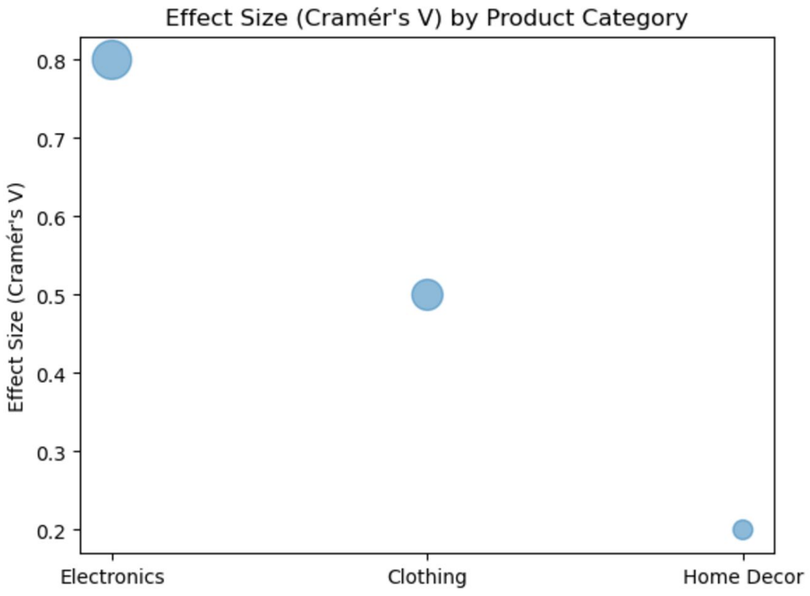
Insights:

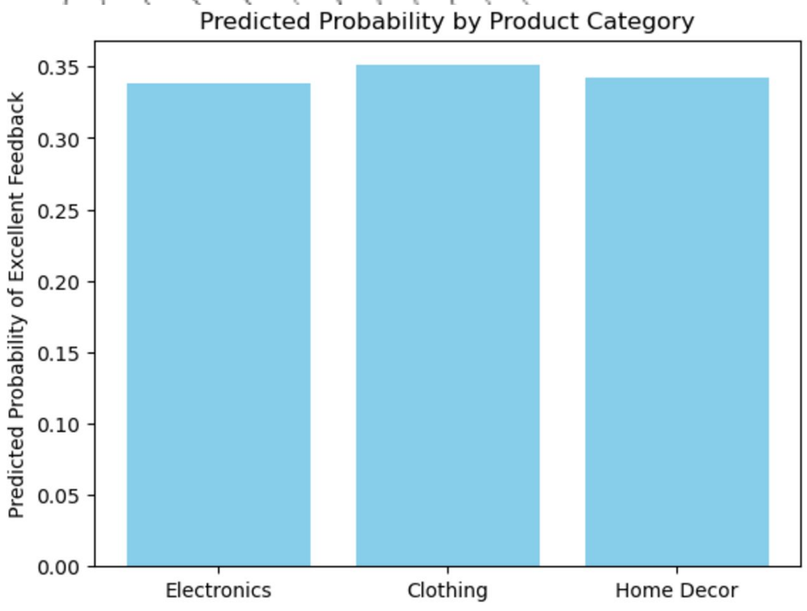
- Demographics (age) show non-linear effects.
- Shipping speed and product categories significantly impact feedback.

Implications:

- Highlights importance of shipping speed

Visualization





Box Plot: Feedback Scores by Shipping Method:

- Customers using "Same-Day" shipping tend to give slightly higher feedback scores compared to "Standard" shipping, reflecting its positive impact.

Bubble Chart: Effect Size by Product Category

- "Same-Day" shipping has the strongest effect on electronics (Cramér's V = 0.8), followed by clothing (0.5), and the weakest for home decor (0.2).

Bar Chart: Predicted Probability of Excellent Feedback

- Predicted probabilities of "Excellent" feedback are similar across product categories, with electronics showing a slight edge over clothing and home decor.

Insights

Conlusion

- Shipping methods influence customer feedback differently across product categories. "Same-Day" shipping improves "Excellent" ratings significantly for time-sensitive categories like **Electronics** and **Clothing**. However, the effect is minimal for **Home Decor**, where shipping speed is less critical. Overall, the impact of "Same-Day" delivery, while measurable, remains modest across categories.

- **Shipping Strategies by Product Category:**
  - **Electronics and Clothing:** Prioritize "Same-Day" delivery to boost satisfaction in time-sensitive segments.
  - **Home Decor:** Focus on improving cost-efficiency and quality, as delivery speed has a smaller influence on feedback.

Insights

Conclusion

- More specifically on the targeted customers
- Enhance consumers' experience
- Online and offline channel integration (inventory)

Age significantly affects purchase amount

Different genders have different preferences for product types and different spending behaviors

Shipping methods have a major influence on customer feedback.

THANK'S FOR WATCHING

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